SS-PLRQR: Super-Spectrum Parallel Low-Rank Quaternion Recovery for Hyperspectral Image Classification

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***Abstract*— Low-rank tensor recovery (LRTR)-based feature extraction from authentic hyperspectral images (HSIs) has become widely employed to improve classification performance by removing sparse errors while preserving multidimensional structures. However, these techniques operate on either individual planar slices or unfolded matrix of each dimension, thereby overlooking spectral cross-channel correlation. Although quaternion-based low-rank recovery takes advantage of spectral cross-channel correlations, it often depends on spectral compression of global spectral bands, overlooking most spectral information, including the heterogeneity of various spectral regions and the homogeneity of regional spectrums. To address the challenges, this article proposes a super-spectrum parallel low-rank quaternion recovery (SS-PLRQR) model composed of a hybrid super-spectrum grouping strategy (HSSGS) and a parallel low-rank quaternion recovery (PLRQR) algorithm. HSSGS streamlines the algorithmic flow and provides necessary conditions for subsequent adaptive analysis by visually partitioning global bands into heterogeneous subregions. Following this, PLRQR effectively and independently eliminates noisy errors from each super-spectrum, utilizing fit parameter values tailored to the heterogeneity of different super-spectrums. It maintains the complete planar spatial structure and cross-channel correlation of homogeneous spectrums within the same quaternionic super-spectrum. Comparative experiments on three real-world HSIs demonstrate the proposed model's remarkable applicability, effectiveness, and robustness to various noises in classification.**

***Index Terms*— Feature extraction, hyperspectral image (HSI), low-rank recovery, quaternion representation.**

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NOMENCLATURE

Symbol	Meaning	Type
$\mathbf{H}$	Hyperspectralimage.	Tensor.
k	Homogeneous spatial patch size.	Scalar.
$\boldsymbol{p}_{\alpha,\beta}$	(α, β)th pixel vector.	Vector.
$\boldsymbol{G}_{\alpha,\beta,b}$	Hybrid spectral–spatial graph.	Matrix.
$\mathbf{A}$	Edge subset of heterogeneous groups.	Matrix.
$\hbar_n$	nth super-spectrums.	Vector.
$\mathbf{\Gamma}_n$	nth homogeneous spectral cluster.	Tensor.
$\mathbf{K}_{n_\vartheta}$	ϑthhomogeneous spectrum.	Matrix.
$\mathbf{\Gamma'}_n$	Updated super-spectrum of $\mathbf{\Gamma}_n$.	Tensor.
$\dot{\boldsymbol{\Gamma}}_n$	Quaternion matrix changed from $\mathbf{\Gamma'}_n$.	Matrix.
$\dot{\boldsymbol{L}}_n$	Low-rank quaternion part of $\dot{\boldsymbol{\Gamma}}_n$.	Matrix.
$\dot{\boldsymbol{N}}_n$	Sparse noisy quaternion from $\dot{\boldsymbol{\Gamma}}_n$.	Matrix.
$\dot{\boldsymbol{\Gamma}}_n^*$	Complex adjoint matrix of $\dot{\boldsymbol{\Gamma}}_n$.	Matrix.
$\dot{\boldsymbol{\Gamma}}_n^{-*}$	Inverse quaternion of $\dot{\boldsymbol{\Gamma}}_n^*$.	Matrix.
λ_n	Regularization parameter.	Scalar.
$\mathring{\boldsymbol{S}}_n^{d+1}$	$d+1$th singular value matrix.	Matrix.
abs$(\cdot)$	Absolute value of the input elements.	Operation.
SVD$(\cdot)$	Quaternion singular value decomposition.	Operation.
$.*$	Multiplication operations on arrays.	Operation.
$./$	Division operation on arrays.	Operation.
length$(\cdot)$	Number of eigenvalues satisfying criteria.	Operation.

I. INTRODUCTION

HYPERSPECTRAL images (HSIs) commonly capture a wealth of spatial–spectral tridimensional information spanning hundreds of continuous visible–infrared spectrums and numerous spatial pixels from observed areas [1], [2].

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Depending on the identifiable differences in reflection characteristics of compositional elements or compounds, various HSI classification approaches can precisely recognize terrestrial, marine, or atmospheric-cover objects lacking artificial category labels [3], [4]. On the one hand, the profuse features broaden the application scope and enhance the reliability of HSI in practical applications [5]. On the other hand, the collection and processing of HSIs are prone to interference from uncertainties, resulting in errors that can cause unconformable observations and deteriorate the performance and stability of standard classifiers [6].

HSI classification has attracted numerous researchers striving for low-rank recovery and extracting essential features from corrupted data, aiming to mitigate the adverse impacts of sparse noise on the utilization of residual features [7]. The fundamental goal of low-rank recovery is to break down the original data into a low-rank component and an extraneous component. Low-rank matrix recovery (LRMR) is an exemplary instance of such algorithms. Initially, principal component analysis (PCA) is designed to divide the initial matrix into a low-rank part and a Gaussian noise part by singular value decomposition (SVD) [8]. Nonlinear PCA [9], Segmented PCA [10], and Superpixelwise PCA [11] trod on the heels of the standard PCA to adaptively prompt the performance of HSI classification approaches. However, the actual noise part of most HSIs may not follow the distribution subject to the given conditions.

To solve these issues, the LRMR technique known as robust PCA (RPCA) [12], [13] is efficiently applied by solving the matching principal component pursuit and thus removing various types of disruptive noise from digital images. The technique is further refined to ensure its adaptability to actual HSIs through leveraging lexicographic reconstruction [14], Moreau-enhanced total variation regularization [15], and adaptive weight graph total variation [16], all of which work in conjunction with LRMR to enhance the quality of the input to the assigned classifiers. LRMR can also be optimized by directly adjusting its norm types without significant morphological change. For instance, the combined nuclear and ℓ_1-norm is tailored to enhance low-rank approximation [17], whereas ℓ_0-norm is employed to transform the standard minimization objective into the mixed integer programming problem [18]. However, the above enhancements are bound to compromise the planar spatial structure of HSI by stretching it into a unidimensional vector quantity. Consequently, it may result in the loss of crucial multidimensional structural information, leading to unsatisfying performance.

Unlike LRMR techniques, employing low-rank tensor recovery (LRTR) on corrupted triaxial data has gained popularity in HSI classification for maintaining multidimensional structures while effectively eliminating sparse errors. LRTR derived from decomposition is primarily carried out through canonical polyadic decomposition [19], Tucker decomposition [20], [21], or tensor SVD (T-SVD) [22], [23]. Canonical polyadic decomposition typically involves transforming the triaxial data into the sum of several rank-one tensors, disregarding the contextual information among different factors. In contrast, tucker LRTR operates by unfolding planarization along each dimension to obtain the core tensor. It retains only the linear correlation of diverse characteristics and disrupts the structure of the other dimensions [24]. The T-SVD LRTR, also known as tensor RPCA (TRPCA), globally achieves the minimized objective function iteratively. It separates the corrupted 3-D HSI into a clean low-rank tensor and a sparse tensor [25]. Similar to LRMR, T-SVD LRTR can be enhanced through the incorporation of additional techniques, such as gradient maps [26], tensor dictionary learning [27], and locality-preserving graphs [28]. Their objective functions can also be customized by adjusting norm regularization. For instance, multilinear TRPCA (M-LRTR) replaces the tensor norm on the low-rank module with a multilinear rank-based total tensor norm and utilizes $\ell_{1,1,2}$-norm to regulate the module of sparse noises [29]. Dual ℓ_p quasi-norm and weighted Schatten p-norm are introduced to restrict the sparse errors of ℓ_p-based TRPCA (P-LRTR) [30] and the clean tensor of the enhanced TRPCA (E-LRTR) [31], respectively. Similarly, nonconvex TRPCA (NN-LRTR) [32], improved TRPCA (I-LRTR) [33], and novel TRPCA (N-LRTR) [34] are adapted models suitable for authentic multidimensional images. Unlike canonical polyadic and Tucker decomposition methods, T-SVD-based techniques remain more essential bidimensional features in corresponding planar slices through block circulation along any dimension than unfolding and rank-one-based LRTR approaches [35].

Although T-SVD-based LRTR can effectively preserve bidimensional features through planar slicing, it compromises the contextual information of the residual variables. In particular, horizontal and lateral slices maintain spatial–spectral structure but disrupt the contextual information of spatial rows and columns, respectively [36], [37]. Holding the natural function of preserving the complete 2-D spatial structure, frontal slices cannot retain spectral context and overlook spectral homogeneity and heterogeneity [29], where homogeneous spectral channels display significantly stronger correlations than heterogeneous ones. Some researchers have confirmed that preserving spatial and spectral contextual information is crucial for data analysis [11], [38]. Furthermore, independently processing data subsets under spectral homogeneity and heterogeneity can effectively and efficiently enhance the quality of purificatory tensors [39].

Fortunately, the quaternion is customarily employed in low-rank recovery for color images to maintain spectral cross-channel correlations [40]. Theoretically, by converting the three spectral bands of a color image into an imaginary part matrix, preserving more spectral contextual and bidimensional spatial structure is feasible than T-SVD-based LRTR using slices [41], [42]. However, quaternionic transformation cannot be directly conducted on HSIs because quaternions accommodate up to four spectral channels. Although the extended version of quaternions, which is hyper-complex, seems to retain most spectral information through a low-rank recovery technique, the corresponding hypercomplex algebra system

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is not yet fully developed. It lacks the primary conditions for practical HSI applications [43]. Therefore, current relative research must incorporate quaternion-based low-rank recovery into HSI using spectral compression techniques such as feature extraction [44] and band selection [45]. Unfortunately, the spectral compression techniques make it difficult to recognize the diversity of heterogeneous spectrum variables and preserve most spectral features.

Another technique to reduce the influence of the noisy part of HSIs is to facilitate the common deep-learning network-based LRTR of grayscale or RGB images to adaptive variants of HSIs [46]. However, not all deep learning networks can jointly consider both the spatial and the spectral correlations, such as convolutional neural networks (CNNs) [47], [48]. To solve this issue, SDe-CNN [49] is designed to denoise the adjacent central band within a certain quantity of a spectral group. However, such a network still ignores the heterogeneity of spatial pixels or spectral bands. Moreover, the generalization has not been fully verified, and they seem unstable in certain situations [50].

Above all, several problems need to be solved. First, most LRTR techniques operate on either individual planar slices or unfolded matrices of each dimension, failing to retain the structure and context of 2-D spatial pixels and spectral bands simultaneously. Second, although quaternion-based low-rank recovery can capture more spectral cross-channel correlations than most LRTR techniques for hyperspectral classification, it commonly relies on compression on global spectral bands, ignoring the heterogeneity of different spectral regions and abandoning most spectral features. This article proposes a super-spectrum parallel low-rank quaternion recovery (SS-PLRQR) model composed of a hybrid super-spectrum grouping strategy (HSSGS) and a parallel low-rank quaternion recovery (PLRQR). The detailed contributions of SS-PLRQR are demonstrated as follows.

1) HSSGS comprises a hybrid spectral–spatial graph representation and a super-spectrum grouping strategy. The reorganized hybrid spectral–spatial graph can efficiently and naturally create a proper bidimensional graph for further spectral heterogeneity recognition. The super-spectrum grouping strategy assigns spectral channels with the same grouping label to the same group, streamlining the algorithm steps and preserving more spectral features. Even if the same band may occur in different band groups, the grouping method retains sufficient information to support subsequent processing and analysis, laying a solid foundation for the subsequent quaternion representation. Meanwhile, the spectral homogeneity and heterogeneity are sufficiently leveraged.
2) PLRQR is comprised of a homogeneous quaternion representation and a parallel recovery algorithm. By converting heterogeneous super-spectrum tensors into different quaternionic matrixes, homogeneous quaternion representation overcomes the challenge of high-dimensional features in HSI that cannot be directly transformed into quaternions. With fit parameter values adapted to spectrum heterogeneity, the parallel recovery algorithm can generate ideal low-rank quaternion components, eliminating useless elements independently and adaptively.

The rest is demonstrated as follows. The related work is presented in Section II. Section III methodically illustrates the proposed SS-PLRQR with detailed descriptions and explanations. Section IV demonstrates and discusses the results of various comparative experiments. Finally, Section V provides the conclusion.

II. Related Work

This section briefly introduces several representative models concerning visualized homogeneity recognition and quaternion representation on HSIs.

A. Visualized Homogeneity Recognition on HSIs

Homogeneity recognition on spatial regions of HSIs is in widespread employment because surface materials are commonly continuous in real-world observation areas, which can be verified in the phenomenon that the same land-cover category labels tend to cluster [51]. In computer vision, superpixels define perceptually uniform regions of a planar image where intracluster pixels are similar in color, brightness, or texture, and adjacent superpixels generally differ [11]. Latterly, superpixel segmentation algorithms have been broadly employed in the homogeneity recognition on hyperspectral datasets by operating on one component of the original spectrums, being capable of delineating the boundaries of heterogeneous pixels pretty quickly and accurately while also providing a distinct visualization of the segmentation outcomes [39]. Jiang et al. [11] merged entropy rate superpixel segmentation with PCA, extracting low-rank features within different homogeneous regions to improve HSI classification accuracy and efficiency. Inspired by this, Li et al. [52] and Jia et al. [51] simultaneously adopted superpixel segmentation to acquire spatial distribution information and enhance performance. Typically, superpixel segmentation supports the convenient and efficient development of regional and complementary spatial information for HSI classification. Nevertheless, the visualized homogeneity recognition on spectral regions of HSIs is still unfashionable and limited by the condition of processing only on the planer graph even though significant similarity exists among adjacent spectral bands holding continuous wavelength [38].

B. Quaternion Representation on HSIs

Quaternion representation can naturally protect the spatial structure and the cross-channel correlations among up to four spectral bands occupying a real part and three imaginary parts [40]. Therefore, this representation is more suitable for color images than HSIs unless compression is performed on spectral dimension. To be adaptive to HSIs, quaternion representation is always merged with band selection or feature extraction while damaging the original spectral inherent structure, most essential information, and the

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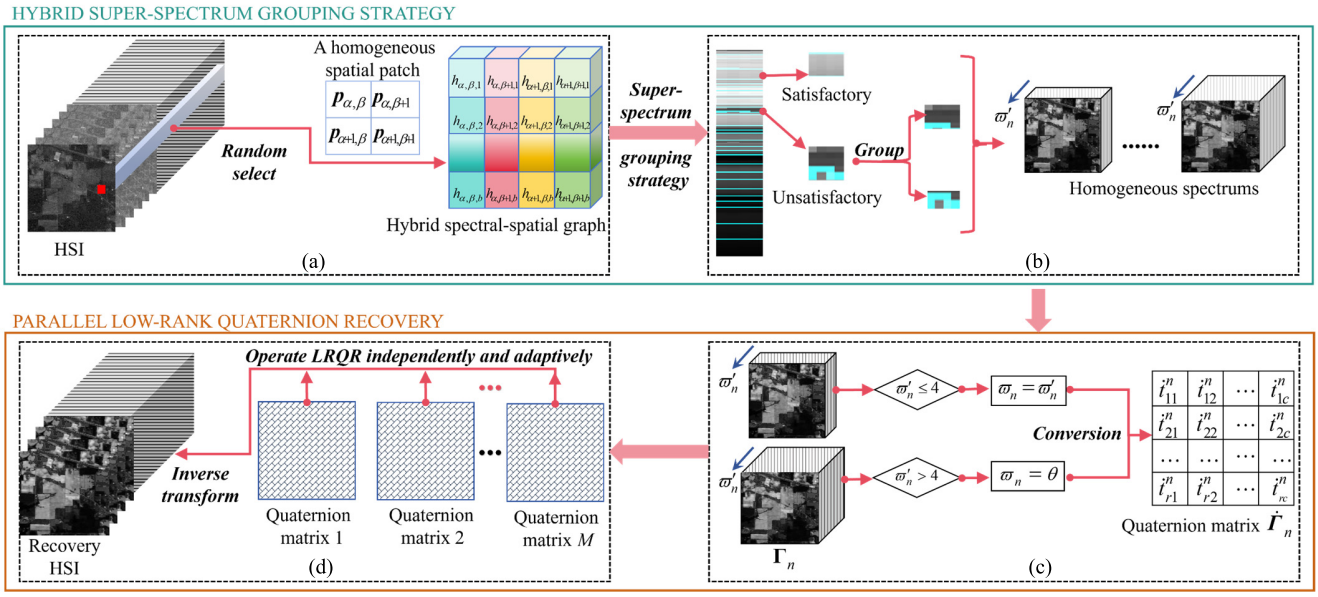


Fig. 1. Flowchart of the proposed SS-PLRQR. (a) Hybrid spectral–spatial graph construction. (b) Super-spectrum grouping strategy. (c) Homogeneous quaterion representation. (d) Parallel recovery algorithm.

contextual and homogeneous information of vast spectrums. In [53], the quaternionic representation is established after performing an extended morphological attribute profile and PCA, which compressed the high-dimensional HSI into only three features to construct a quaternion CNN. Similarly, Li et al. [45] established a quaternion Weber local descriptor histogram on the first three principal components of the input hyperspectral datasets by PCA. Although there is insufficient study on the effect of quaternion representation on HSIs, relative research has already firmly proven that quaternionic transformation plays an essential role in denoising traditional color images [40], [41], [42]. One of the challenges that needs to be addressed is how to protect most spectral characteristics and structures when executing quaternion representation on HSIs.

III. Method

In this article, bold straight letters denote 3-D tensors, bold capital italic letters indicate planar matrices, bold lowercase italic letters represent vectors, and italic letters denote scalars, as demonstrated in the Nomenclature. The flowchart of the proposed SS-PLRQR is shown in Fig. 1. Specifically, HSSGS is conducted on a reorganized hybrid spectral–spatial graph including random homogeneous spatial patch, producing heterogeneous spectrum clusters for further designed quaternion representation. Subsequently, PLRQR independently and in parallel operates on distinct heterogeneous super-spectrums to obtain the final low-rank features for HSI classification.

A. Hybrid Super-Spectrum Grouping Strategy

1) Hybrid Spectral–Spatial Graph Construction: To ensure homogeneity recognition, it is essential to construct a planar graph in the visualized homogeneity recognition-related state of the arts. However, the current planer graphs designed for super-pixel recognition degrade spectral information utilization and are at odds with the objective of homogeneous super-spectrum recognition [39]. To address this issue, this article first develops a novel reconstructive hybrid spectral–spatial graph where grouping homogeneous spectrums becomes feasible and practical. Before establishing the objective graph, a proper homo-class labeled spatial patch needs to be randomly selected, which is expected to be highly unformed and similar to eliminate the impact of spatial variables and highlight spectral differences as much as possible. In theory, as the scope of the test area gradually expands, the distance between the objects represented by the selected pixels also increases, leading to a decreasing trend in their overall similarity. In other words, adding more pixel points can increase disorganization in spectrum segmentation.

The process starts with a random initial point, and additional pixels are gradually included by expanding the spatial patch outward along the increasing Euclidean distance from this initial point. The construction of this spatial patch only considers the pixels in the lower right direction of the initial point. When the Euclidean distance is the same, the pixel to the right of the initial point is included in the spatial patch earlier than the point below. Meanwhile, the initial pixel should be reset when the category labels of selected pixels are inconsistent [54]. For example, the pixel selection order is shown by the number on each square (pixel point) in Fig. 2(a), where the square marked with the number 1 is the initial pixel in the spatial patch. In Fig. 2(a), yellow and green squares represent pixels labeled with two different classes, while white means the class of the pixel is unknown. To reflect the impact of patch size k, the segmentation outcomes with qualitative green heterogeneous boundaries are displayed under increasing k in Fig. 2(b). From the results, the heterogeneity curves commonly get bumpier as the value of k increases, with more and more protrusions or depressions and

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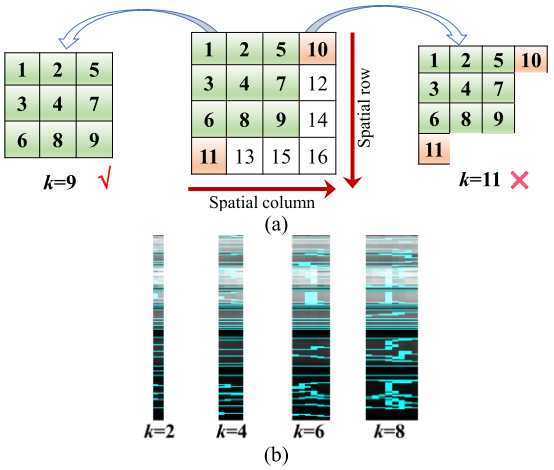


Fig. 2. (a) Selection of the ideal spatial patch. (b) Segmentation outcomes with qualitative green boundaries under increasing k.

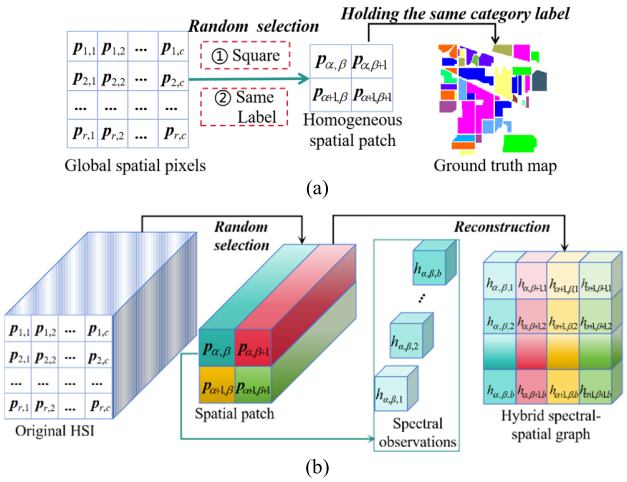


Fig. 3. (a) Random square homogeneous spatial patch selection. (b) Construction of hybrid spectral–spatial graph.

TABLE I

CLASSIFICATION ACCURACIES UNDER RANDOM PATCHES WITH SPATIAL SIZE CHOSEN FROM {2, 3, 4, 5, 6}

Dataset	Index	k=2	k=3	k=4	k=5	k=6
Indian Pines	1	98.18	98.57	98.20	98.68	98.64
	2	97.96	98.70	98.06	98.44	97.97
	3	98.09	98.32	98.11	**98.85**	***98.77***
	4	98.12	98.46	98.03	98.49	98.51
	5	98.25	98.42	98.12	98.26	98.28
Pavia University	1	***96.34***	95.92	96.02	94.88	95.76
	2	95.14	94.96	95.61	95.08	94.87
	3	94.20	95.91	95.79	95.29	95.52
	4	95.18	95.74	95.76	96.14	94.77
	5	94.51	**97.37**	96.10	94.82	94.18
Longkou	1	97.66	97.52	97.50	97.69	97.52
	2	97.41	97.51	97.24	97.59	97.73
	3	97.41	97.48	97.42	97.47	97.74
	4	97.61	97.55	**98.21**	97.43	***97.78***
	5	97.70	97.61	97.29	97.34	97.61

irregular curves between different spectrums. Fewer samples must be selected to ensure the chosen pixels have greater consistency.

Subsequently, to further explore the specific impact of the randomness of the spatial patch on the proposed model and provide a reference for determining an appropriate small patch size, Table I demonstrates model performance under five independently and randomly selected patches of fixed spatial pixel size k within {2, 3, 4, 5, 6}. The suboptimal values in Table I are marked in italics to be distinguished from the optimal values in bold. In Table I, the optimal and suboptimal results of the model on different experimental datasets usually occur at different patch sizes and without significant regularity. Since the patches are randomly selected, the sizes of these values do not reflect the random advantages of the model under different patch sizes. Still, they provide reference conditions for model formulation.

Specifically, the size and random selection of patches within a specific range will not significantly impact the experimental results. Still, the growing size of spatial patches increases the computational burden. It makes the visualization of the initial super-spectrum grouping unsatisfactory, according to Fig. 2. It can be found that for all datasets, the classification results in $k = 4$ are relatively stable. Moreover, the shape of the spatial patch under this setting is somewhat regular. Meanwhile, the position distance between pixels is relatively compact, and more pixels may increase the variability of the segmentation boundaries. In a nutshell, the growing size of spatial patches can increase the computational burden and make the visualization of the initial super-spectrum grouping unsatisfactory. Therefore, the setting $k = 4$ is a proper option for SS-PLRQR. While a patch size $k = 4$ is appropriate for applying hyperspectral remote sensing datasets that have remarkable adjacent spectral consistency and neighboring pixel similarity, it may not be universally proper and generalizable for all applications of different types of HSIs. For other remote sensing datasets, the authors suggest that when other labeled samples are scattered, or the image of the ground objects is irregular, the patch size k can be appropriately reduced to 2, 3, or 4.

Suppose that the original HSI $\mathbf{H} \in \mathbb{R}^{r\times c\times b}$ with the entries $h_{i,j,t}(i = 1, 2, \ldots, r; j = 1, 2, \ldots, c; t = 1, 2, \ldots, b)$ can be denoted as $\{\boldsymbol{p}_{\alpha,\beta}|\alpha = 1, 2, \ldots, r; \beta = 1, 2, \ldots, c\}$. Fig. 3(a) illustrates the sufficient conditions for selecting a random homogeneous square spatial patch holding four adjacent and homo-labeled pixels. Then, the observed values of all spectral variables on a square homogeneous spatial patch composed of adjacent home-category pixel vectors $\{\boldsymbol{p}_{\alpha,\beta}, \boldsymbol{p}_{\alpha,\beta+1}, \boldsymbol{p}_{\alpha+1,\beta}, \boldsymbol{p}_{\alpha+1,\beta+1}\}$ with spatial raw position α and column position β, i.e., the objective hybrid spectral–spatial graph $\boldsymbol{G}_{\alpha,\beta,b}$ can be represented as

$$\boldsymbol{G}_{\alpha,\beta,b} = \begin{pmatrix} h_{\alpha,\beta,1} & h_{\alpha,\beta+1,1} & h_{\alpha+1,\beta,1} & h_{\alpha+1,\beta+1,1} \\ h_{\alpha,\beta,2} & h_{\alpha,\beta+1,2} & h_{\alpha+1,\beta,2} & h_{\alpha+1,\beta+1,2} \\ \vdots & \vdots & \vdots & \vdots \\ h_{\alpha,\beta,b} & h_{\alpha,\beta+1,b} & h_{\alpha+1,\beta,b} & h_{\alpha+1,\beta+1,b} \end{pmatrix}. \quad (1)$$

The corresponding visual construction of the objective hybrid spectral–spatial graph $\boldsymbol{G}_{\alpha,\beta,b}$ is displayed in Fig. 3(b). The proposed grouping strategy introduces homogeneous spatial patches with the same category labels rather than exploring

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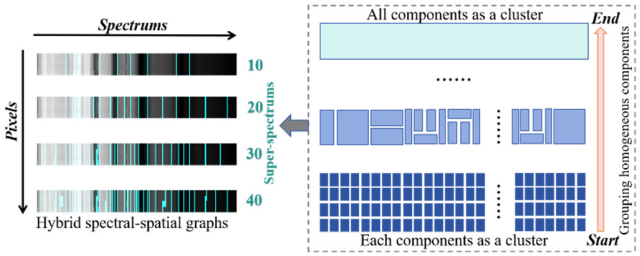


Fig. 4. Initial super-spectrum grouping hierarchy. The figure demonstrates the initial grouping with 10, 20, 30, and 40 super-spectrums and the visual grouping process from individual clusters to the global cluster.

optimal regions in pixel spaces holding homo-category samples [54]. This approach ensures superior reproducibility and applicability to various datasets and helps preserve more spectral features.

2) Super-Spectrum Grouping Strategy: An entropy rate-based super-spectrum grouping strategy produces clusters after representing the hybrid spectral–spatial graph. Given the vertex set $\boldsymbol{\Upsilon}$ and the edge set $\boldsymbol{E}$, the hybrid spectral–spatial graph $\boldsymbol{G}_{\alpha,\beta,b}$ can be segmented into M connected uniform subsets by finding a proper edge subset $\boldsymbol{A} \subseteq \boldsymbol{E}$. The initial objective function consisting of the entropy rate term $R(\cdot)$ and balancing term $B(\cdot)$ can be acquired by optimizing the edge subset $\boldsymbol{A}$ until meeting the following conditions [11]:

$$\underset{\boldsymbol{A}\subseteq\boldsymbol{E}}{\operatorname{argmax}}(\operatorname{tr}(R(\boldsymbol{A})) + \chi B(\boldsymbol{A})). \tag{2}$$

Here, trace function $\operatorname{tr}(\cdot)$ and $\chi > 0$ refer to the sum of the diagonal elements of the input matrix and the parameter to make a balance between two different terms, respectively. Benefiting from the entropy rate term $R(\cdot)$ producing compact and homogeneous subregions and balancing term $B(\cdot)$ regulating similar super-spectrum sizes, the preliminary grouping labels are generated. Fig. 4 presents the initial super-spectrum grouping hierarchy on the Indian Pines dataset, with edges that indicate heterogeneity marked in green.

It should be noted that the grouping processing is intended to acquire a super-spectrum of a specific size. This construction method treats global bands as variables and replaces spectral vectors with reconstructive bidimensional graphs, preserving the physical meaning of the original observations. Nevertheless, from a visual perspective, the initial grouping of homogeneous spectrums often leads to visually uneven boundaries, with multiple protrusions or depressions and irregular curves spanning various spectrums. This may be due to the high similarity across nonadjacent bands in a cross-channel manner or the consequence of the distributions of selected pixels not being entirely consistent.

The proposed model abandons additional forced operations to generate smooth, straight boundaries between heterogeneous spectral groups [54]. Instead, it relies solely on the results of the super-spectrum segmentation to perform spectral band grouping. Specifically, it assigns each spectrum to corresponding groups based on the super-spectrum labels of different homogeneous pixels within the same band, which means that a single band may appear in different homogeneous groups and nonadjacent spectral bands may be merged into one cluster due to its constituent pixels probably having different super-spectrum labels. The super-spectrums $\hbar_n (n = 1, 2, \ldots, M)$ on the graph $\boldsymbol{G}_{\alpha,\beta,b}$ may overlap

$$\begin{aligned}\boldsymbol{G}_{\alpha,\beta,b} &= \bigcup_{n=1}^{M} \bigcup_{\vartheta=1}^{\varpi_n'} \left\{h_{\alpha,\beta,n_\vartheta}, h_{\alpha,\beta+1,n_\vartheta}, h_{\alpha+1,\beta,n_\vartheta}, h_{\alpha+1,\beta+1,n_\vartheta}\right\} \\ &= \bigcup_{n=1}^{M} \hbar_n \text{ s.t. } \bigcup_{n=1}^{M} \bigcup_{\vartheta=1}^{\varpi_n'} n_\vartheta = \{1, 2, \ldots, b\}\end{aligned} \tag{3}$$

where $n_\vartheta (\vartheta = 1, 2, \ldots, \varpi_n')$ refers to the index of ϑth spectral band holding in the nth super-spectrum $\hbar_n$ and ϖ_n' denotes the number of spectrums in $\hbar_n$. On account of the super-spectrums of the graph $\boldsymbol{G}_{\alpha,\beta,b}$, the nth homogeneous spectral cluster $\boldsymbol{\Gamma}_n$ is denoted as

$$\boldsymbol{\Gamma}_n = \left\{\boldsymbol{K}_{n_1}, \boldsymbol{K}_{n_2}, \ldots, \boldsymbol{K}_{n_{\varpi_n'}}\right\}, \text{ s.t. } \bigcup_{n=1}^{M} \boldsymbol{\Gamma}_n = \mathbf{H}. \tag{4}$$

Here, $\boldsymbol{K}_{n_\vartheta}$ is the ϑth homogeneous spectrum in the cluster $\boldsymbol{\Gamma}_n$ and the n_ϑth spectrum matrix of $\mathbf{H}$

$$\boldsymbol{K}_{n_\vartheta} = \begin{pmatrix} h_{1,1,n_\vartheta} & h_{1,2,n_\vartheta} & \ldots & h_{1,c,n_\vartheta} \\ h_{2,1,n_\vartheta} & h_{2,2,n_\vartheta} & \ldots & h_{2,c,n_\vartheta} \\ \ldots & \ldots & \ldots & \ldots \\ h_{r,1,n_\vartheta} & h_{r,2,n_\vartheta} & \ldots & h_{r,c,n_\vartheta} \\ \cdot & & & \end{pmatrix}. \tag{5}$$

The homogeneous spectral band clusters $\boldsymbol{\Gamma}_n (n = 1, 2, \ldots, M)$ represent the final super-spectrums. So far, the model has successfully designed an HSSGS to overcome the limitation of the adjacency of homogeneous bands. It also preserves more spectral context information and simplifies the algorithm process of the model. While this flexible grouping method increases data redundancy, it enhances the algorithm's adaptability to different datasets and random homogeneous spatial patches. Even if the same band may occur in different band groups, the grouping method retains sufficient information to support subsequent processing and analysis, laying a foundation for subsequent quaternion representations. The number M of super-spectrums can be appropriately adjusted according to the research objectives. The number M of super-spectrums can be reduced for spectral variables with significant wavelength differences and small numbers. The number M of super-spectrums can be appropriately increased for datasets with a large number of spectral bands.

B. Parallel Low-Rank Quaternion Recovery

To fully preserve the homogeneous spectral contextual and cross-channel information while preserving spatial structural features during low-rank recovery, this study introduces PLRQR on each super-spectrum independently.

1) Homogeneous Quaternion Representation: Quaternion representation is customarily employed on color images to protect cross-channel correlations. Inspired by its construction principle and advantages, this article designs the homogeneous quaternion representation approach to safeguard the bidimensional spatial structure, essential context, and cross-channel correlations of homogeneous spectral bands. To implement

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quaternion representation within each homogeneous band group, the number of elements within any $\mathbf{\Gamma}_n$ must meet the specific condition $\varpi_n' \leq 4$ [40].

Theoretically, the HSSGS generates homogeneous spectral subsets $\mathbf{\Gamma}_n (n = 1, 2, \ldots, M)$ with similar volumes as much as possible, using the entropy rate term $R(\cdot)$ and balancing term $B(\cdot)$. However, in some super-spectrums, the number of spectral variables ϖ_n' exceeds the standard upper limit. It should be noted that increasing the number of homogeneous spectrum clusters cannot avoid generating excess spectral numbers. While high-dimensional hypercomplex arrays can address this issue, the transformation and computation of related algorithms still need to be fully integrated and can bring significant computational burdens. Therefore, the proposed model must adopt additional spectral quantity normalization methods to create conditions for subsequent computations. Specifically, this model remains top θ components of super-spectrums containing over four spectrums by PCA. Therefore, the updated spectrum number ϖ_n of the nth super-spectrum $\mathbf{\Gamma}_n$ is

$$\varpi_n = \begin{cases} \varpi_n' & \varpi_n' \leq 4 \\ \theta & \varpi_n' > 4 \end{cases} \tag{6}$$

where the number of remaining components θ is no more than four $\theta \leq 4$. Correspondingly, the preceding entries $h_{i,j,t}$, spectrum matrix $\boldsymbol{K}_{n_\vartheta} (\vartheta = 1, 2, \ldots, \varpi_n')$, and super-spectrums $\mathbf{\Gamma}_n$ are transformed into $h'_{i,j,t}$, $\boldsymbol{K}'_{n_\vartheta} (\vartheta = 1, 2, \ldots, \varpi_n)$, and $\mathbf{\Gamma}'_n$, respectively. Each super-spectrum tensor $\mathbf{\Gamma}'_n (n = 1, 2, \ldots, M)$ can be approximately reconstructed into a quaternion matrix $\dot{\boldsymbol{\Gamma}}_n$ as follows:

$$\dot{\boldsymbol{\Gamma}}_n = \begin{bmatrix} \dot{t}^n_{11} & \dot{t}^n_{12} & \cdots & \dot{t}^n_{1c} \\ \dot{t}^n_{21} & \dot{t}^n_{22} & \cdots & \dot{t}^n_{2c} \\ \cdots & \cdots & \cdots & \cdots \\ \dot{t}^n_{r1} & \dot{t}^n_{r2} & \cdots & \dot{t}^n_{rc} \end{bmatrix} = \left(\dot{t}^n_{ij}\right)_{r \times c}. \tag{7}$$

Here, the entries $\dot{t}^n_{ij} (i = 1, 2, \ldots, r;\ j = 1, 2, \ldots, c)$ of the nth matrix $\dot{\boldsymbol{\Gamma}}_n \in \mathbb{Q}^{r \times c}$ refer to complex or hypercomplex numbers of under-limited size. Due to such transformation, the spatial structure is well-maintained in the bidimensional quaternion matrix. Meanwhile, the spectral context and cross-channel correlation information on each spatial pixel are preserved through complex or hypercomplex representation [42]. To be more specific, the entry $\dot{t}^n_{ij}$ can be represented as

$$\dot{t}^n_{ij} = \begin{cases} h'_{i,j,n_1}\mu & \varpi_n = 1 \\ h'_{i,j,n_1}\mu + h'_{i,j,n_2}\nu & \varpi_n = 2 \\ h'_{i,j,n_1}\mu + h'_{i,j,n_2}\nu + h'_{i,j,n_3}\delta & \varpi_n = 3 \\ h'_{i,j,n_1}\mu + h'_{i,j,n_2}\nu + h'_{i,j,n_3}\delta + h'_{i,j,n_4} & \varpi_n = 4 \end{cases} \tag{8}$$

where μ, ν, and δ represent the basis complex operators that satisfy the following conditions [41]:

$$\begin{aligned} &\mu^2 = \nu^2 = \delta^2 = \mu\nu\delta = -1 \\ &\mu\nu = -\nu\mu = \delta,\ \nu\delta = -\delta\nu = \mu,\ \delta\mu = -\mu\delta = \nu. \end{aligned} \tag{9}$$

As a result, the model outputs M quaternion matrices that contain planar spatial structure, essential context, and cross-channel correlations of homogeneous spectral bands. The proposed model overcomes the challenge of high-dimensional features in HSI that cannot be directly transformed into quaternions suitable for color images with a maximum of four channels. This is achieved by recognizing homogeneous super-spectrum tensors and converting them into quaternionic matrixes. As a result, subsequent PLRQR becomes more convenient.

2) Parallel Recovery Algorithm: This article presents a PLRQR model to maximize the dispersion of quaternion elements composed of homogeneous spectral bands. Under the constraint of least-squares error, maximizing the scattering of projected quaternion samples is equivalent to minimizing the reconstruction error between the projected quaternion samples and the original samples, thus ensuring the accuracy and stability of data recovery. Suppose $\sigma_i(\cdot)$ is the ith singular value of the input matrix and $|\cdot|$ denotes the modulus of the input quaternion matrix, the quaternion trace norm on $\dot{\boldsymbol{\Gamma}}_n$ is $\|\dot{\boldsymbol{\Gamma}}_n\|_{\mathrm{N}} = \sum_i \sigma_i(\dot{\boldsymbol{\Gamma}}_n)$, the quaternion $\ell_{2,1}$-norm is $\|\dot{\boldsymbol{\Gamma}}_n\|_{2,1} = \sum_{i=1}^{r} (\sum_{j=1}^{c} |\dot{t}_{ij}|^2)^{1/2}$, the quaternion infinite norm is $\|\dot{\boldsymbol{\Gamma}}_n\|_\infty = \max|\dot{t}_{ij}|$, the quaternion ℓ_2-norm is $\|\dot{\boldsymbol{\Gamma}}_n\|_2 = \max(\sigma_i(\dot{\boldsymbol{\Gamma}}_n))$, and the quaternion Frobenius-norm is $\|\dot{\boldsymbol{\Gamma}}_n\|_{\mathrm{F}} = (\sum_i (\sigma_i(\dot{\boldsymbol{\Gamma}}_n))^2)^{1/2}$. Therefore, the objective function of PLRQR on the nth matrix $\dot{\boldsymbol{\Gamma}}_n$ derived from the corresponding homogeneous super-spectrum $\boldsymbol{\Gamma}'_n$ can be defined as [40]

$$\min_{\dot{\boldsymbol{L}}_n, \dot{\boldsymbol{N}}_n} \left(\|\dot{\boldsymbol{L}}_n\|_N + \lambda_n \|\dot{\boldsymbol{N}}_n\|_{2,1}\right), \text{ s.t. } \dot{\boldsymbol{\Gamma}}_n = \dot{\boldsymbol{L}}_n + \dot{\boldsymbol{N}}_n \tag{10}$$

where $\lambda_n = ((\max(r, c))^{1/2}\varpi_n)^{-1}$ represents the regularization parameter adopted to balance the two different norms. $\dot{\boldsymbol{L}}_n$ denotes the finial optimal low-rank quaternion component obtained after parallel data transformation and $\dot{\boldsymbol{N}}_n$ refers to the abandoned sparse noisy quaternion from $\dot{\boldsymbol{\Gamma}}_n$. To achieve the best possible result for the objective function, this model employs an augmented lagrange multiplier method within the alternating direction method of multipliers (ADMM) framework [34], [44]. As a result, this approach transforms the objective function, which does not have a closed-form solution, into a constrained optimization problem. On the strength of the basis quaternion operators μ, ν, and δ, the quaternion matrix $\dot{\boldsymbol{\Gamma}}_n$ can be expressed by transformed homogeneous spectrums $\boldsymbol{K}'_{n_\vartheta} (\vartheta = 1, 2, \ldots, \varpi_n)$ as

$$\dot{\boldsymbol{\Gamma}}_n = \begin{cases} \boldsymbol{K}'_{n_1}\mu & \varpi_n = 1 \\ \boldsymbol{K}'_{n_1}\mu + \boldsymbol{K}'_{n_2}\nu & \varpi_n = 2 \\ \boldsymbol{K}'_{n_1}\mu + \boldsymbol{K}'_{n_2}\nu + \boldsymbol{K}'_{n_3}\delta & \varpi_n = 3 \\ \boldsymbol{K}'_{n_1}\mu + \boldsymbol{K}'_{n_2}\nu + \boldsymbol{K}'_{n_3}\delta + \boldsymbol{K}'_{n_4} & \varpi_n = 4. \end{cases} \tag{11}$$

Above all, $\dot{\boldsymbol{\Gamma}}_n$ indeed can be regarded as a quaternion function $\boldsymbol{K}'_{n_1}\mu + \boldsymbol{K}'_{n_2}\nu + \boldsymbol{K}'_{n_3}\delta + \boldsymbol{K}'_{n_4}$ containing at least one value from $\{\boldsymbol{K}'_{n_4}, \boldsymbol{K}'_{n_3}, \boldsymbol{K}'_{n_2}\}$ equal to zero. In addition, $\dot{\boldsymbol{\Gamma}}_n$ can be represented as $\dot{\boldsymbol{\Gamma}}_n = \dot{\boldsymbol{\Psi}}_a + \dot{\boldsymbol{\Psi}}_b\nu$ where $\dot{\boldsymbol{\Psi}}_a = \boldsymbol{K}'_{n_4} + \boldsymbol{K}'_{n_1}\mu$ and $\dot{\boldsymbol{\Psi}}_b = \boldsymbol{K}'_{n_2} + \boldsymbol{K}'_{n_3}\mu$. The conjugate complex number of $\dot{\boldsymbol{\Gamma}}_n$ is $\bar{\boldsymbol{\Gamma}}_n = \boldsymbol{K}'_{n_4} - (\boldsymbol{K}'_{n_1}\mu + \boldsymbol{K}'_{n_2}\nu + \boldsymbol{K}'_{n_3}\delta)$; thus, the matching complex adjoint matrix $\dot{\boldsymbol{\Gamma}}^*_n$ is denoted as [41]

$$\dot{\boldsymbol{\Gamma}}^*_n = \begin{bmatrix} \dot{\boldsymbol{\Psi}}_a & \dot{\boldsymbol{\Psi}}_b \\ -\bar{\dot{\boldsymbol{\Psi}}}_b & \bar{\dot{\boldsymbol{\Psi}}}_a \\ \cdot & \end{bmatrix}. \tag{12}$$

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To demonstrate the operating process of the PLRQR model, Algorithm 1 illustrates the actual execution flow on each quaternion $\dot{\boldsymbol{\Gamma}}_n (n = 1, 2, \ldots, M)$ methodically. In Algorithm 1, iter denotes the upper limit of iterations for the PLRQR model within each quaternion. $(\hat{\boldsymbol{S}}_n^{d+1} - 1/\upsilon_n^d)_+$ represents a diagonal matrix that shrinks the eigenvalues of $\hat{\boldsymbol{S}}_n^{d+1}$ that are less than $1/\upsilon_n^d$ to zero. It should be noted that the convergence conditions are decided following [40] and [41], indicating that the algorithm would be terminated immediately when the approximation error $\|\dot{\boldsymbol{\Gamma}}_n - \dot{\boldsymbol{L}}_n^{d+1} - \dot{\boldsymbol{N}}_n^{d+1}\|_2$ minus the influence of itself $\|\dot{\boldsymbol{\Gamma}}_n\|_{\mathrm{F}}$ is less than the set value ε or the number of iterations reaches the upper limit iter. This study iteratively updates until the objective function converges, generating the final low-rank quaternion component that eliminates sparse noise while preserving the complete planar spatial structure and cross-channel and contextual information among homogeneous spectrums in parallel.

Algorithm 1 PLRQR on $\dot{\boldsymbol{\Gamma}}_n (n = 1, 2, \ldots, M)$

Input: $\dot{\boldsymbol{\Gamma}}_n, \varpi_n, \rho = 1.5, \varepsilon = 10^{-7}, iter = 100, r, c$
Output: low-rank tensor $\mathbf{L}$
For $n = 1, 2, \ldots, M$ **do**
1: Initialize $d = 1, \dot{\boldsymbol{N}}_n^d = \mathbf{0} \in \mathbb{R}^{r \times c}$
$\lambda_n = (\sqrt{\max(r, c)}\varpi_n)^{-1}, \upsilon_n^d = 1.25/\|\dot{\boldsymbol{\Gamma}}_n\|_2$
$\dot{\boldsymbol{Y}}_n^d = \dot{\boldsymbol{\Gamma}}_n / \max(\|\dot{\boldsymbol{\Gamma}}_n^*\|_2, \lambda_n^{-1}\|\dot{\boldsymbol{\Gamma}}_n\|_\infty)$
While not converged do
2: Obtain $\dot{\boldsymbol{L}}_n^{d+1} = \arg\min_{\dot{\boldsymbol{L}}_n} (\|\dot{\boldsymbol{L}}_n\|_{\mathrm{N}} + \frac{\upsilon_n^d}{2}\|\dot{\boldsymbol{L}}_n + \dot{\boldsymbol{N}}_n^d - \dot{\boldsymbol{\Gamma}}_n + \frac{\dot{\boldsymbol{Y}}_n^d}{\upsilon_n^d}\|_{\mathrm{F}})$
2.1: $[\hat{\boldsymbol{U}}_n^{d+1}, \hat{\boldsymbol{S}}_n^{d+1}, \hat{\boldsymbol{V}}_n^{d+1}] = SVD(\dot{\boldsymbol{\Gamma}}_n - \dot{\boldsymbol{N}}_n^d + \dot{\boldsymbol{Y}}_n^d/\upsilon_n^d)^*$
2.2: $l_n = length(\hat{\boldsymbol{S}}_n^{d+1} - 1/\upsilon_n^d)_+$
2.3: $\hat{\boldsymbol{W}}_n^{d+1} = \hat{\boldsymbol{U}}_n^{d+1}(:, 1 : l_n)(\hat{\boldsymbol{S}}_n^{d+1} - 1/\upsilon_n^d)_+(\hat{\boldsymbol{V}}_n^{d+1}(:, 1 : l_n))^{\mathrm{T}}$
2.4: $\dot{\boldsymbol{L}}_n^{d+1} = (\hat{\boldsymbol{W}}_n^{d+1})^{-*}$
3: Obtain $\dot{\boldsymbol{N}}_n^{d+1} = \arg\min_{\dot{\boldsymbol{N}}_n} (\|\dot{\boldsymbol{N}}_n\|_{2,1} + \frac{\upsilon_n^d}{2}\|\dot{\boldsymbol{L}}_n^{d+1} + \dot{\boldsymbol{N}}_n - \dot{\boldsymbol{\Gamma}}_n + \frac{\dot{\boldsymbol{Y}}_n^d}{\upsilon_n^d}\|_{\mathrm{F}})$
3.1: $\dot{\boldsymbol{N}}_n^{d+1} = \dot{\boldsymbol{\Gamma}}_n - \dot{\boldsymbol{L}}_n^{d+1} + \dot{\boldsymbol{Y}}_n^d/\upsilon_n^d$
3.2: $\dot{\boldsymbol{N}}_n^{d+1} = \max(1 - \frac{\lambda_n}{\upsilon_n^d}./abs(\dot{\boldsymbol{N}}_n^{d+1}), 0). * \dot{\boldsymbol{N}}_n^{d+1}$
4: Update $\dot{\boldsymbol{Y}}_n^{d+1} = \dot{\boldsymbol{Y}}_n^d + \upsilon_n^d(\dot{\boldsymbol{\Gamma}}_n - \dot{\boldsymbol{L}}_n^{d+1} - \dot{\boldsymbol{N}}_n^{d+1})$
5: Update $\upsilon_n^{d+1} = \rho\upsilon_n^d$
6: Compute $d = d + 1$
7: Until one of the following conditions is satisfied
$\|\dot{\boldsymbol{\Gamma}}_n - \dot{\boldsymbol{L}}_n^{d+1} - \dot{\boldsymbol{N}}_n^{d+1}\|_2/\|\dot{\boldsymbol{\Gamma}}_n\|_{\mathrm{F}} < \varepsilon$ or $d \geq iter$
End while
8: Transform optimal low-rank quaternions $\dot{\boldsymbol{L}}_n$ into real 3-D array tensors $\mathbf{L}_n$ by the inverse function of (11).
End For
9: Construct the final low-rank tensor $\mathbf{L}$ by $\mathbf{L} = \bigcup_{n=1}^{M} \mathbf{L}_n$

C. Complexity Analysis

This section briefly analyzes the computational complexity of the proposed SS-PLRQR on an input HSI $\mathbf{H} \in \mathbb{R}^{r \times c \times b}$. First, it costs $O(rc)$ to select a proper square patch to build (1) at most. After that, an initial super-spectrum segmentation in [55] is conducted on the established spectral–spatial graph holding kb. Benefiting from very few updates performed at each iteration, the complexity of (2) approximates $O(kb\log(kb))$. Then, the designed finial super-spectrum grouping cost nearly $O(kb)$ to acquire (3)–(5). After the HSSGS stage displayed above, PLRQR begins with the normalization by PCA to acquire ϖ_n in (6), which leads to $O(rc\sum_{n=1}^{M}\varpi_n'^2)$. Then, the operation of PLRQR on $\dot{\boldsymbol{\Gamma}}_n (n = 1, 2, \ldots, M)$ in (7) is similar to its in [40], as detailed in Algorithm 1. Specifically, the computational complexity generated for calculating λ_n is $O(r^2c^2\varpi_n)$ and for calculating (10) is $O(cr^2 + r^3\varpi_n Q_n)$, where Q_n is the actual times of alternating minimization iteration on $\dot{\boldsymbol{\Gamma}}_n$. In a nutshell, the complexity of SS-PLRQR is $O(kb\log(kb) + r^2c^2\sum_{n=1}^{M}\varpi_n + r^3\sum_{n=1}^{M}\varpi_n Q_n)$.

IV. Experiments and Analyses

A. Hyperspectral Datasets and Experimental Setup

The experiments primarily utilize three publicly available hyperspectral datasets, which are Indian Pines [56], Pavia University [57], and WHU-HI-Longkou (Longkou) [58], [59]. These datasets are notable for their extensive range of spectral bands and have been widely used in research related to hyperspectral imaging.

1) *Indian Pines:* An airborne AVIRIS sensor acquired this HSI within a specific scene on June 12, 1992. After abandoning 24 water-absorption channels, the HSI holds 200 available spectral bands with a wavelength range between 400 and 2500 nm. Each spectral band contains 145 × 145 pixels (10 249 labeled samples), which can be artificially divided into 16 categories.
2) *Pavia University:* A ROSIS sensor gathered this HSI in Pavia, Italy. It contains 103 spectral bands and 610 × 340 spatial pixels (42 776 labeled samples), including nine material categories.
3) *Longkou:* It was gathered by a unmanned aerial vehicle (UAV) platform equipped with a hyperspectral sensor. The available data cube consists of 270 spectral channels with a resolution of 0.463 m. Additionally, the dataset contains 550 × 400 pixels (204 542 labeled samples) covering nine crops.

To handle a large number of spatial pixels and spectral bands, 10% of labeled pixels from each class in Indian Pines, 2% of labeled pixels from each class in Pavia University, and 5% randomly selected labeled pixels from each class in Longkou were used to form the training sets for the corresponding classifiers. The test set comprised the remaining labeled pixels. All experiments underwent fivefold cross-validation to average the results, except for parameter tuning, which required only one classification. A multiclass support vector machine (SVM) classifier is employed to assess the performance of low-rank recovery approaches to HSI classification [60]. The classifier is equipped with the radial basis function kernel containing parameters C and γ, which can be acquired through cross-validation as $C = 10$ and $\gamma = 0.125$ on the three datasets.

To ensure an objective comparison, the SVM classifier's training sets and parameter settings were identical, and each cross-validation on the same hyperspectral remote sensing image was fixed rather than random. This study adopted widely used classification accuracy metrics, i.e., the accuracy of each category (CA), averaging accuracy (AA) of each category, overall accuracy (OA), and kappa coefficient

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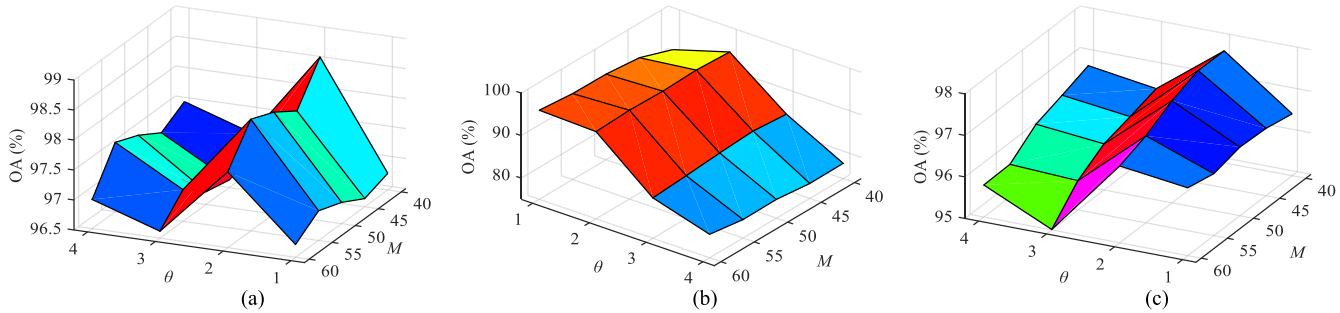


Fig. 5. Parameter tuning results in OA (%) on (a) Indian Pines, (b) Pavia University, and (c) Longkou.

(Kappa), to assess the quality of features obtained by the comparative algorithms. All experiments are conducted in MATLAB R2023b on a laptop with a 2.60-GHz CPU, Windows 11 system, Intel Core i9-13900H processors, and 32-GB memory.

B. Parameter Tuning

To establish the whole model and obtain competitive results, fine-tuning parameters are necessary for further comparison. The proposed SS-PLRQR model relies on two critical parameters, i.e., the number of homogeneous super-spectrums M in (3) obtained through the super-spectrum grouping strategy and the number of remaining components $\theta(\theta \leq 4)$ in (6) to maintain feature consistency for homogeneous quaternion representation. By analyzing the effect of these parameters on the model's performance, the performance of SS-PLRQR can be further improved. To convert each homogeneous super-spectrum tensor into a quaternion matrix holding no more than four variables, the value range of the parameter θ is $\{1, 2, 3, 4\}$ across different hyperspectral datasets. Meanwhile, considering the enormous spectral volume, the parameter M for the Indian Pines, Pavia University, and Longkou images is chosen from $\{40, 45, 50, 55, 60\}$. Fig. 5 demonstrates the performance of SS-PLRQR under different parameter settings on three HSIs.

In Fig. 5, each vertex's coordinates on the subplots represent the combined effect of M and θ, where a higher vertex value indicates better model performance under that combined parameter setting. On the Pavia University dataset, OA decreases as θ increases, indicating a relatively high sensitivity to the change of this parameter. In contrast, the OA is less affected by the value of M, suggesting that the SS-PLRQR exhibits a certain degree of robustness to different M values on the Pavia University dataset. On Pavia University and Longkou, the model consistently performs best when $(M, \theta) = (45, 2)$. Unlike the Pavia University data, the OA on the Longkou exhibits an insignificant trend with different settings of M and θ, reflecting that the inherent differences in spectral and spatial structures between different HSIs may remarkably affect model performance. Similarly, to achieve the best classification performance, the recommended parameter setting on Indian Pines is $(M, \theta) = (40, 2)$.

When comparing the results, it is clear that our model consistently achieves optimal performance across different hyperspectral datasets when $\theta = 2$. Since θ is closely related to the construction of quaternion representation, the optimal performance can be attributed to the fact that our model under this setting provides sufficient nonredundant spectral cross-channel information for low-rank quaternion recovery. This study has successfully optimized the model to achieve its best performance through parameter tuning. It supports subsequent parallel comparisons with other methods and is an essential reference for parameter selection in practical applications on different real-world HSIs.

C. Robust Classification Performance to Different Datasets

Several relevant algorithms are adopted as comparative methods to demonstrate the reliability of the proposed model to HSI classification. Since the proposed SS-PLRQR is derived from quaternion representation and low-rank recovery, the comparison methods include the LRMR-based RPCA [13], low-rank complex recovery-based complex RPCA (CRPCA) [43], and LRTR methods, i.e., M-LRTR [29], P-LRTR [30], E-LRTR [31], NNTRPC [32], N-LRTR [34], and SPRB-TRPCA [39]. Meanwhile, deep learning-based low-rank recovery SDe-CNN [49] and feature extraction RP-Net [47] have also been tested to comprehensively evaluate the robustness and superiority of the proposed method in classification tasks. The results of all methods are directly input into the SVM without any additional processing, and the parameter settings are broadly consistent with the matching literature. The best results are highlighted in bold, and the suboptimal values are marked in underlined italics.

Tables II–IV present the averaging CA, OA, AA, and Kappa values with corresponding standard errors for different low-rank recovery algorithms on the Indian Pines, Pavia University, and Longkou datasets, respectively. In Tables II–IV, the proposed method consistently outperforms other algorithms across all tested HSIs, fully validating its effectiveness, robustness, and applicability. Specifically, on Longkou, the proposed SS-PLRQR achieves the highest OA, AA, and Kappa, demonstrating its excellent classification performance. As shown in Table III, compared to SPRB-TRPCA, the proposed model outperforms it by approximately 3.4% in AA. This significant advantage not only reflects the effectiveness of the quaternion representation algorithm in extracting cross-spectral channel information, but also highlights the applicability of different HSIs.

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TABLE II
CLASSIFICATION ACCURACIES WITH STANDARD DEVIATIONS (AFTER ±) ON INDIAN PINES (%)

Class	RPCA	CRPCA	M-LRTR	P-LRTR	E-LRTR	NN-LRTR	N-LRTR	SPRB-TRPCA	RP-Net	SDe-CNN	SS-PLRQR
Alfalfa	74.84±1.58	62.65±1.22	28.64±0.87	69.94±1.09	85.21±0.91	82.43±0.37	85.93±0.92	**100.0±0.00**	**100.0±0.00**	91.09±0.60	***97.17±0.50***
Co-N-T	74.72±0.22	71.29±0.28	60.14±0.30	76.63±0.22	86.60±0.21	81.30±0.32	87.06±0.33	***95.73±0.11***	89.49±0.38	90.57±0.07	**98.63±0.10**
Co-M-T	79.33±0.21	77.44±0.25	48.18±0.33	79.03±0.21	88.49±0.18	84.28±0.34	89.15±0.09	***95.33±0.12***	94.67±0.12	88.59±0.27	**96.01±0.17**
Corn	58.70±0.48	63.96±0.82	31.63±0.41	63.12±0.40	69.65±0.20	86.91±0.42	68.26±0.08	82.11±0.51	***91.75±0.56***	76.17±0.55	**97.58±0.20**
Grass-P	82.73±0.31	88.34±0.44	43.17±0.30	90.41±0.38	91.68±0.26	86.03±0.80	92.24±0.36	96.73±0.28	***97.98±0.08***	93.21±0.26	**98.18±0.08**
Grass-T	90.11±0.12	90.79±0.06	60.91±0.24	91.17±0.15	95.42±0.08	88.26±0.13	95.10±0.09	**100.0±0.00**	97.39±0.17	96.54±0.14	***99.64±0.02***
Gr-P-M	76.04±1.53	89.49±0.72	23.82±1.41	89.64±0.68	86.27±0.65	74.85±1.98	89.52±0.83	87.13±1.48	***90.53±0.77***	**91.50±0.88**	74.72±2.33
Hay-W	94.98±0.28	91.90±0.13	78.27±0.58	93.66±0.27	97.81±0.11	86.73±0.11	97.31±0.10	99.63±0.06	***99.67±0.02***	98.44±0.07	**100.0±0.00**
Oats	60.45±1.12	49.17±4.51	2.070±0.46	88.51±1.12	81.93±0.69	72.67±1.96	81.71±0.74	**100.0±0.00**	***98.46±0.34***	74.81±1.65	**100.0±0.00**
Soy-N-T	74.12±0.28	76.50±0.21	57.29±0.25	80.11±0.29	87.04±0.23	85.17±0.11	86.79±0.16	***97.15±0.16***	95.89±0.12	91.62±0.09	**97.16±0.21**
Soy-M-T	78.85±0.11	79.74±0.21	62.94±0.15	83.10±0.17	91.38±0.09	86.71±0.19	90.63±0.09	***95.47±0.08***	94.69±0.15	95.14±0.03	**98.76±0.05**
Soy-C-T	77.86±0.32	72.19±0.32	69.77±0.62	72.05±0.32	85.11±0.33	86.61±0.35	83.58±0.28	95.61±0.11	***96.00±0.22***	87.68±0.38	**96.95±0.20**
Wheat	92.38±0.41	94.03±0.36	55.83±0.85	92.04±0.37	99.03±0.12	98.10±0.08	99.35±0.09	98.87±0.25	***99.46±0.12***	99.03±0.10	**100.0±0.00**
Woods	89.73±0.16	91.96±0.12	83.61±0.30	92.63±0.08	95.54±0.13	96.87±0.14	95.51±0.14	99.34±0.10	**99.56±0.04**	98.33±0.11	***99.10±0.08***
Buildings	73.72±0.43	75.06±0.48	66.71±0.50	73.03±0.70	86.53±0.50	93.66±0.29	88.49±0.47	94.29±0.24	***98.60±0.05***	91.10±0.30	**99.02±0.09**
St-S-T	84.84±0.88	98.73±0.07	81.92±1.03	99.12±0.08	99.17±0.08	89.53±0.88	***99.19±0.08***	98.94±0.06	**99.71±0.07**	98.81±0.07	98.72±0.16
OA	80.54±0.06	80.89±0.04	61.90±0.11	83.09±0.07	90.25±0.07	87.28±0.07	90.11±0.03	***96.42±0.04***	95.47±0.07	93.17±0.05	**98.29±0.03**
AA	78.96±0.11	79.58±0.25	53.43±0.16	83.39±0.21	89.18±0.15	86.26±0.26	89.36±0.13	96.02±0.11	***96.49±0.04***	91.41±0.07	**96.98±0.14**
Kappa	77.76±0.06	78.11±0.05	56.43±0.12	80.67±0.07	88.89±0.08	85.47±0.08	88.71±0.03	***95.91±0.05***	94.82±0.08	92.21±0.06	**98.05±0.03**

TABLE III
CLASSIFICATION ACCURACIES WITH STANDARD DEVIATIONS (AFTER ±) ON PAVIA UNIVERSITY (%)

Class	RPCA	CRPCA	M-LRTR	P-LRTR	E-LRTR	NN-LRTR	N-LRTR	SPRB-TRPCA	RP-Net	SDe-CNN	SS-PLRQR
Asphalt	94.88±0.13	95.75±0.03	46.95±0.36	95.91±0.07	96.34±0.06	96.49±0.08	96.30±0.07	***96.63±0.05***	**97.98±0.05**	93.69±0.20	95.96±0.08
Meadows	91.61±0.04	96.97±0.03	77.57±0.31	96.95±0.06	98.21±0.01	97.91±0.04	98.12±0.04	**98.73±0.02**	98.65±0.07	97.65±0.06	***98.48±0.04***
Gravel	60.92±0.31	78.33±0.66	50.64±0.41	77.24±0.50	82.31±0.44	78.83±0.58	80.44±0.61	***83.70±0.43***	82.73±0.03	81.36±0.66	**92.65±0.20**
Trees	88.41±0.37	92.02±0.17	25.12±0.43	96.68±0.06	96.61±0.06	***97.26±0.09***	96.50±0.08	**97.49±0.29**	97.28±0.01	96.01±0.23	81.97±0.29
P-metal	97.85±0.17	99.45±0.12	64.39±0.53	97.14±0.23	97.88±0.09	96.58±0.39	97.30±0.26	97.23±0.19	**99.98±0.00**	98.99±0.08	***99.04±0.07***
Bare Soil	70.19±0.22	84.83±0.22	56.98±0.44	83.77±0.26	87.91±0.16	87.49±0.20	89.11±0.12	***95.47±0.10***	91.20±0.03	89.48±0.16	**98.62±0.05**
Bitumen	62.40±0.64	66.92±0.54	40.08±0.29	70.47±0.45	75.73±0.16	74.02±0.40	77.02±0.39	67.72±0.40	***81.08±0.01***	71.79±0.92	**98.25±0.21**
Self-B	78.23±0.34	81.17±0.23	32.98±0.18	80.48±0.36	82.00±0.31	82.61±0.36	81.41±0.18	83.73±0.19	**89.88±0.02**	77.53±0.27	***89.34±0.20***
Shadows	99.96±0.00	99.67±0.03	23.81±0.45	99.93±0.00	***99.96±0.00***	***99.96±0.00***	***99.96±0.00***	***99.96±0.00***	**100.0±0.00**	**100.0±0.00**	94.91±0.22
OA	85.28±0.08	91.39±0.09	57.44±0.10	91.60±0.05	93.42±0.04	93.04±0.09	93.42±0.08	94.43±0.05	***95.33±0.03***	92.33±0.07	**95.67±0.04**
AA	82.71±0.11	88.34±0.15	46.50±0.10	88.73±0.07	90.77±0.06	90.13±0.14	90.68±0.13	91.18±0.07	***93.20±0.06***	89.61±0.10	**94.36±0.03**
Kappa	80.57±0.10	88.64±0.12	44.90±0.16	88.90±0.07	91.30±0.06	90.80±0.12	91.29±0.10	92.68±0.06	***93.82±0.04***	89.83±0.09	**94.24±0.05**

TABLE IV
CLASSIFICATION ACCURACIES WITH STANDARD DEVIATIONS (AFTER ±) ON LONGKOU (%)

Class	RPCA	CRPCA	M-LRTR	P-LRTR	E-LRTR	NN-LRTR	N-LRTR	SPRB-TRPCA	RP-Net	SDe-CNN	SS-PLRQR
Corn	93.11±0.14	98.36±0.02	86.47±0.32	97.76±0.04	98.14±0.02	97.92±0.03	98.04±0.03	98.31±0.02	***99.55±0.02***	99.27±0.03	**99.65±0.03**
Cotton	68.65±0.30	78.31±0.39	77.02±0.27	75.67±0.23	76.73±0.18	76.24±0.16	77.28±0.29	81.80±0.16	***89.69±0.17***	*85.27±0.32*	**98.41±0.06**
Sesame	72.71±0.34	91.75±0.55	42.00±0.87	87.30±0.37	87.68±0.32	87.19±0.30	88.09±0.26	90.65±0.30	96.93±0.34	***98.45±0.15***	**99.80±0.03**
Broad-LS	87.91±0.06	94.60±0.09	86.94±0.36	93.54±0.14	94.71±0.08	94.45±0.10	94.39±0.11	95.54±0.06	95.49±0.06	**97.34±0.08**	***95.94±0.07***
N-L-S	65.56±1.16	67.92±0.39	63.62±0.69	67.42±0.56	68.24±0.47	66.20±0.44	68.92±0.62	***86.37±0.14***	82.56±0.46	82.39±0.62	**93.45±0.32**
Rice	94.37±0.10	97.14±0.14	81.23±0.27	95.95±0.04	96.36±0.07	96.05±0.08	96.52±0.08	98.06±0.06	98.85±0.06	**99.02±0.11**	***99.00±0.04***
Water	99.92±0.01	99.92±0.00	97.64±0.04	99.95±0.00	***99.98±0.00***	99.97±0.00	***99.98±0.00***	***99.98±0.00***	99.91±0.00	**99.99±0.00**	99.65±0.01
Road-H	87.54±0.41	92.38±0.26	71.76±0.20	89.91±0.40	91.78±0.27	90.83±0.27	91.91±0.28	93.35±0.19	**94.83±0.23**	***94.51±0.17***	92.60±0.11
Mixed-W	86.72±0.32	88.80±0.14	51.95±0.98	92.07±0.17	92.77±0.27	91.86±0.12	93.59±0.28	93.15±0.13	**97.79±0.11**	***96.61±0.12***	89.29±0.29
OA	91.78±0.01	95.65±0.03	87.29±0.10	94.98±0.03	95.56±0.02	95.30±0.02	95.54±0.03	96.65±0.02	97.37±0.01	***97.69±0.04***	**97.77±0.01**
AA	84.05±0.12	89.91±0.10	73.18±0.18	88.84±0.07	89.60±0.10	88.97±0.08	89.86±0.09	93.02±0.03	***95.07±0.08***	94.76±0.09	**96.42±0.07**
Kappa	89.11±0.01	94.27±0.04	83.26±0.13	93.37±0.05	94.15±0.03	93.81±0.02	94.12±0.04	95.59±0.02	96.53±0.02	***96.96±0.05***	**97.06±0.02**

As displayed in Table IV, SS-PLRQR outperforms the SPRB-TRPCA in terms of CA in all other categories except for the Water, Road-H, and Mixed-W categories in Longkou. By utilizing its distinctive mathematical structure, the quaternion representation-based algorithm can extract cross-spectral channel information more effectively, helping the model capture more nuanced and crucial features of HSI classification. The proposed model also exhibits excellent CA in most land cover categories. As shown in Table II, the proposed model achieves the best classification results for 11 categories of global 16 classes. Although SPRB-TRPCA, RP-Net, and SDe-CNN achieve the second-best performance on Indian Pines, Pavia University, and Longkou, respectively, they are not as stable and robust as SS-PLRQR. Figs. 6–8 display the corresponding visualization of the classification maps for each algorithm. Fig. 9 employs boxplots to illustrate the distribution of accuracy values over five cross-validations.

As shown in Fig. 9, fluctuating waves exist in the accuracy values of different algorithms across different training sets. Overall, SS-PLRQR exhibits outstanding performance in all cases. Specifically, the model achieves the highest accuracy with a narrow range of the boxplot, indicating that it has a high classification accuracy and demonstrates stable performance. Meanwhile, the subfigures in Figs. 6–8 display

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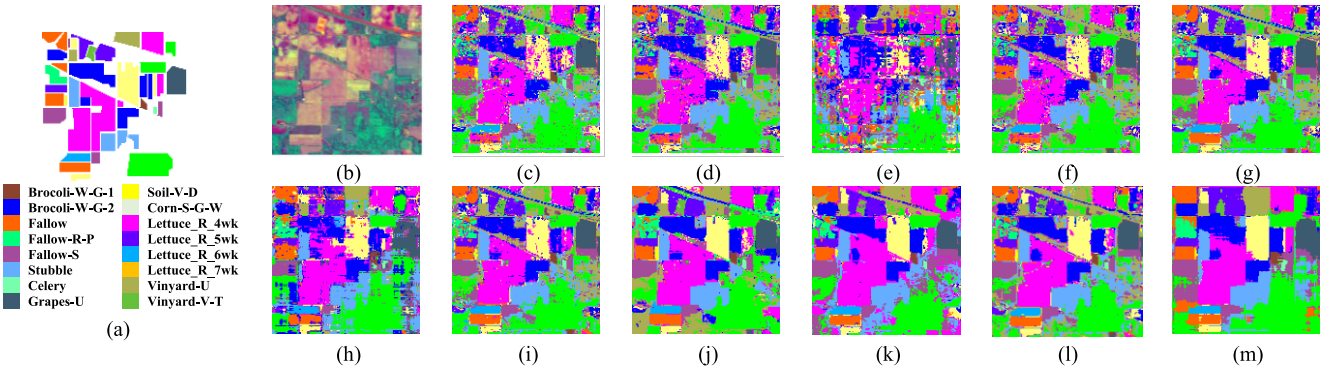


Fig. 6. Classification maps of different comparative algorithms on Indian Pines. (a) Ground-truth labels. (b) False RGB image. (c) RPCA. (d) CRPCA. (e) M-LRTR. (f) P-LRTR. (g) E-LRTR. (h) NN-LRTR. (i) N-LRTR. (j) SPRB-TRPCA. (k) RP-Net. (l) SDe-CNN. (m) SS-PLRQR.

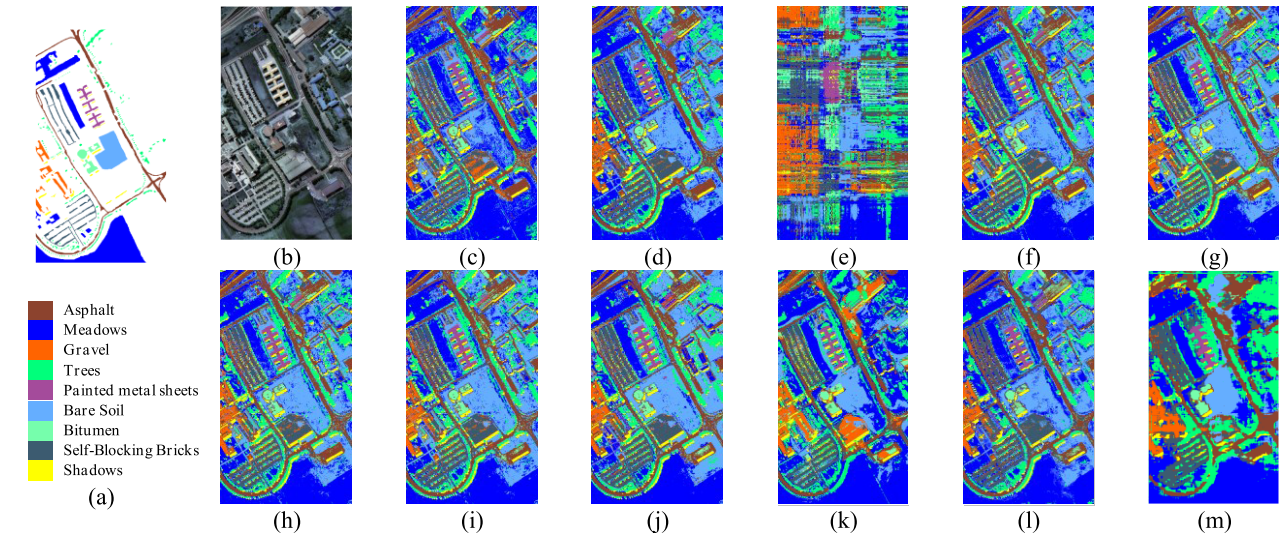


Fig. 7. Classification maps of comparative algorithms on Pavia University. (a) Ground-truth labels. (b) False RGB image. (c) RPCA. (d) CRPCA. (e) M-LRTR. (f) P-LRTR. (g) E-LRTR. (h) NN-LRTR. (i) N-LRTR. (j) SPRB-TRPCA. (k) RP-Net. (l) SDe-CNN. (m) SS-PLRQR.

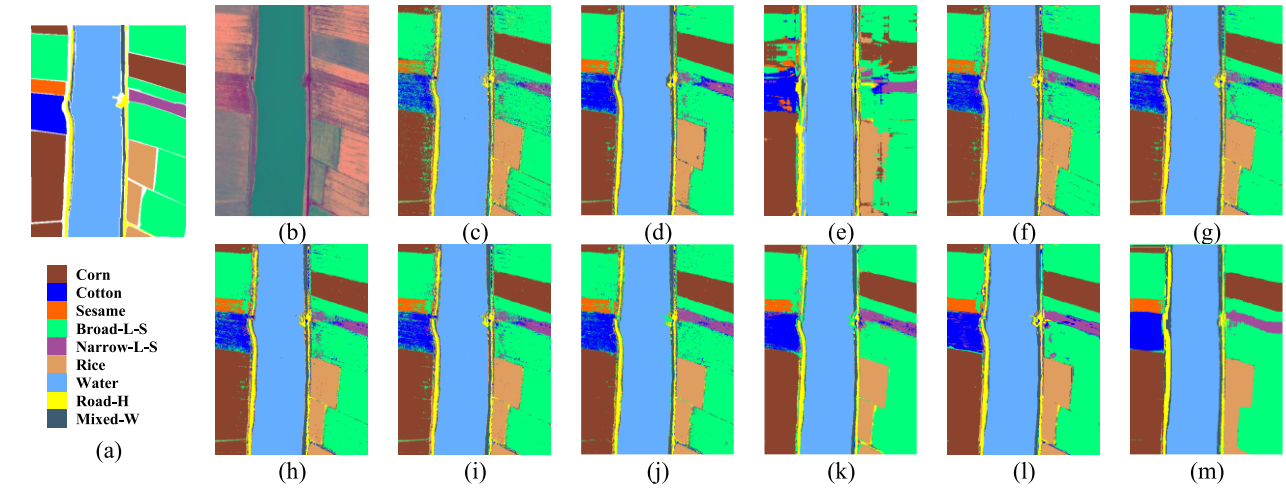


Fig. 8. Classification maps of different comparative algorithms on Longkou. (a) Ground-truth labels. (b) False RGB image. (c) RPCA. (d) CRPCA. (e) M-LRTR. (f) P-LRTR. (g) E-LRTR. (h) NN-LRTR. (i) N-LRTR. (j) SPRB-TRPCA. (k) RP-Net. (l) SDe-CNN. (m) SS-PLRQR.

the corresponding visualized classification maps by each algorithm obtained under the same training and testing pixel sets. Compared to other algorithms, M-LRTR consistently underperforms across all datasets, with inferior results on the Pavia University dataset. This may be due to M-LRTR's reliance on the tensor tubal norm and singular-tubal shrinkage to constrain the sparse component, which overlooks the variance among spectral bands. Additionally, the matching

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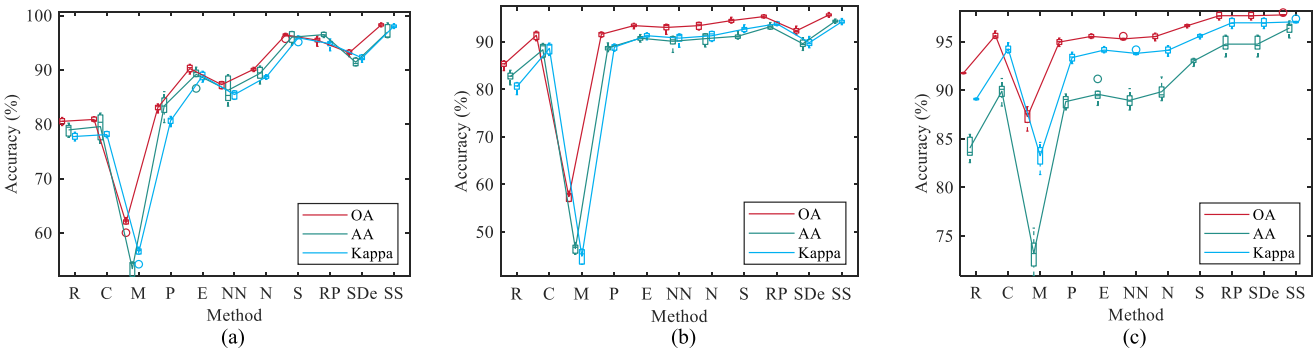


Fig. 9. Boxplots of accuracy (%) acquired by RPCA (R), CRPCA (C), M-LRTR (M), P-LRTR (P), E-LRTR (E), NN-LRTR (NN), N-LRTR (N), SPRB-TRPCA (S), RP-Net (RP), SDe-CNN (SDe), and SS-PLRQR (SS) on (a) Indian Pines, (b) Pavia University, and (c) Longkou.

regularization parameter $\lambda_n = ((\max(r, c, b))^{1/2})^{-1}$ negatively affects the diversity between spatial and spectral elements. The disparity in the sizes of spatial pixels and spectral bands is more pronounced in the Pavia University dataset than in the others. These design limitations in M-LRTR also contribute to the algorithm's prolonged convergence time. Fortunately, the proposed algorithm addresses these deficiencies effectively, which can be seen in the comparison of running time. On the contrary, the classification results obtained by the proposed model show a significant reduction in the number of misclassified pixels, with a more concentrated distribution and significantly fewer noisy pixel points. This results in a pixel classification visualization closer to the actual ground object category labels, thereby verifying the excellent performance of the proposed model in eliminating sparse noise interference.

D. Simulated Experiments

In this section, simulated experiments are conducted by adding noise to the original HSIs, aiming to explore the robustness of SS-PLRQR in noisy situations. The following four cases are considered since those noises usually contaminate HSIs.

1) *Case 1*: The independent and identically distributed Gaussian noise of $\sigma_n = 5$ is added to each spectrum of the normalized HSI.
2) *Case 2*: The independent and identically distributed Gaussian noise of $\sigma_n = 10$ is added to each spectrum of the normalized HSI.
3) *Case 3*: The random sparse noise with the level of 30% is added to the normalized dataset [34].
4) *Case 4*: The setting of random sparse noise is the same as in case 3. Furthermore, the independent and identically distributed Gaussian noise of $\sigma_n = 5$ is also added to each spectrum of HSIs.

Table V displays the classification results obtained by SVM on Indian Pines destroyed as cases 1–4 by the proposed method and the other techniques of the same type. The best results are highlighted in bold, and the suboptimal values are marked in bold italics. It is worth noting that SS-PLRQR always yields superior results over the compared methods, similar to the performance of the original HSI. The proposed approach is pretty anti-interference and is robust to different Gaussian noise, random sparse noise, and mixed noise. Since the suboptimal results are not always obtained by one algorithm and the classifier fails entirely sometimes (marked with underlines), other methods are not as exceptionally stable as SS-PLRQR.

E. Ablation Study

The ablation experiment aims to validate the effectiveness of the two primary modules of the proposed SS-PLRQR: HSSGS and PLRQR. This study conducts comparative classification experiments on the original hyperspectral datasets (all bands), LRTR model (TR), HSSGS-based TR (HB-TR) where PLRQR is replaced with TR, superpixelwise adjacent band grouping based PLRQR (SA-PL) where module HSSGS is replaced with the homogeneous band grouping process in [54], and the proposed model (Our). Fig. 10 illustrates the classification results in averaging OA, AA, and Kappa on three employed hyperspectral datasets. In Fig. 10, the classification accuracy on different datasets gradually improves with the continuous optimization of the model. First, the accuracy observations of TR consistently surpass those of all bands, thereby validating the effectiveness of LRTR.

Subsequently, HB-TR and SA-PL can significantly improve classification accuracy compared to the results through TR. This phenomenon reflects the individual promotions of HSSGS in extracting spectral heterogeneity and of PLRQR in removing noisy parts in hyperspectral classification tasks. Although the trend of performance improvement is insignificant in the Pavia University image, SS-PLRQR performs exceptionally well on other datasets, still verifying the effectiveness of HSSGS in the proposed model.

F. Running Time and Computational Complexity

Due to the employment of the augmented Lagrange multiplier within the ADMM framework, most LRTR models tend to have longer runtimes compared to traditional denoising models on $\mathbf{H} \in \mathbb{R}^{r \times c \times b}$. To intuitively reflect the efficiency of SS-PLRQR, Table VI and Fig. 11 display the complexity and runtime of different algorithms on three HSIs, respectively. The number ζ of iterations of ADMM is denoted for all globally operating models. The quantity of local iterations

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TABLE V

CLASSIFICATION ACCURACIES OF ALGORITHMS OF THE SAME TYPE FACING VARIOUS SIMULATED CASES ON INDIAN PINES (%)

Case	Metric	RPCA	CRPCA	M-LRTR	P-LRTR	E-LRTR	NN-LRTR	N-LRTR	SPRB-TRPCA	RP-Net	SDe-CNN	SS-PLRQR
Case 1	OA	64.49	***96.57***	58.18	55.21	55.20	75.23	62.42	90.20	80.77	95.93	**97.35**
	AA	59.79	95.01	48.38	52.06	52.23	74.37	55.51	93.11	68.68	***95.37***	**97.03**
	Kappa	59.04	***96.09***	51.77	46.83	46.81	71.47	56.00	88.82	77.40	95.36	**96.98**
Case 2	OA	57.00	***93.42***	59.88	49.96	49.93	70.18	52.72	85.30	56.81	89.48	**95.57**
	AA	51.49	***90.30***	49.90	29.70	29.69	74.40	39.04	87.53	61.84	89.87	**96.56**
	Kappa	49.83	***92.46***	54.05	39.37	39.36	65.35	43.32	83.20	46.59	87.95	**94.95**
Case 3	OA	40.56	0.90	44.09	23.96	23.96	81.12	76.13	***91.01***	23.96	86.85	**94.30**
	AA	23.38	0.06	35.10	1.50	1.50	80.50	74.93	89.18	1.50	***90.83***	**94.95**
	Kappa	28.17	0.00	33.91	0.00	0.00	78.40	72.61	***89.73***	0.00	84.88	**93.49**
Case 4	OA	40.54	0.90	44.40	23.96	23.96	23.96	52.85	80.71	23.96	***85.53***	**98.57**
	AA	18.98	0.06	36.06	1.50	1.50	1.49	40.66	77.97	1.50	***89.86***	**98.37**
	Kappa	27.54	0.00	33.55	0.00	0.00	0.00	43.58	77.92	0.00	***83.38***	**98.37**

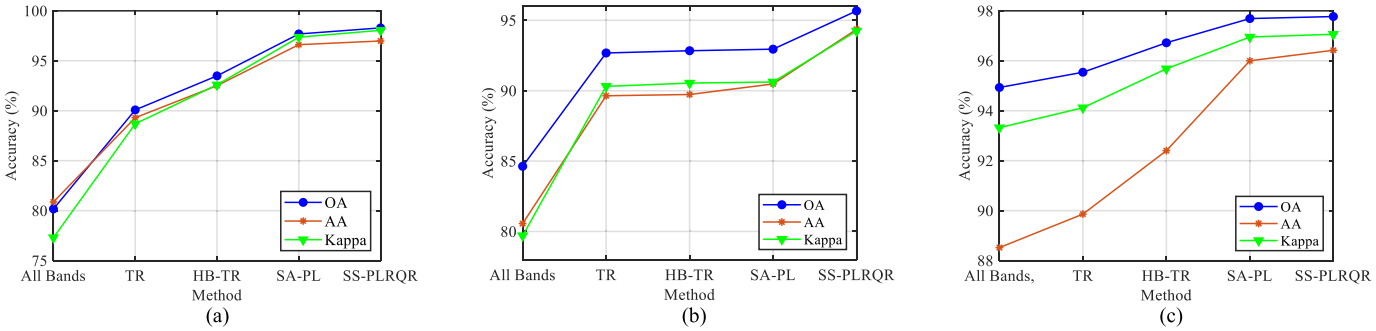


Fig. 10. Ablation experimental classification results (%) in averaging OA, AA, and Kappa obtained by all bands, TR, HB-TR, SA-PL, and SS-PLRQR on (a) Indian Pines, (b) Pavia University, and (c) Longkou.

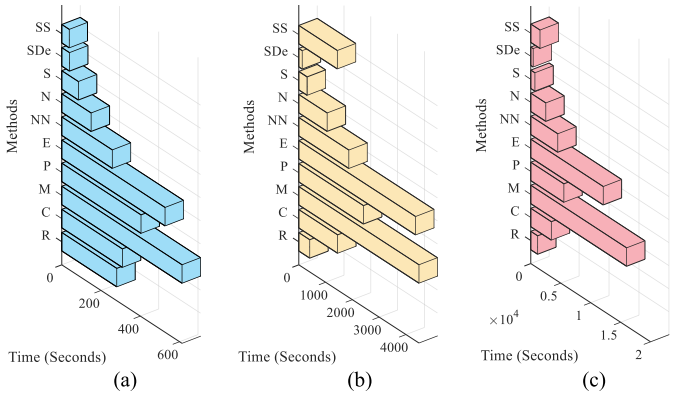


Fig. 11. Computing time (s) of different methods on (a) Indian Pines, (b) Pavia University, and (c) Longkou.

TABLE VI

COMPUTATIONAL COMPLEXITY OF EACH MODEL

Model	Computational Complexity
RPCA	$O(\zeta rcb^2)$
CRPCA	$O(r^2cb + k\zeta_1\zeta_2 r^3)$
M-LRTR	$O(\zeta(rcb\log(b) + \kappa_1\vartheta_1^2 rcb)$
P-LRTR	$O(\zeta(rcb + rcb\log(b) + \kappa_1\vartheta_1^2 b))$
E-LRTR	$O(\zeta(rcb\log(b) + (\vartheta_1 rcb^2)^p)$
NN-LRTR	$O(\zeta(rcb\log(b) + \vartheta_1 rcb^2))$
N-LRTR	$O(\zeta(rcb\log(b) + \kappa_1\vartheta_1^2 b))$
SPRB-TRPCA	$O(\zeta'(mb\log(n) + N_{(1)}N_{(2)}^2 n))$
SDe-CNN	$O(rcK^2\sum_{i=1}^{b}\zeta_i)$
SS-PLRQR	$O(kblog(kb) + r^2c^2\sum_{n=1}^{M}\varpi_n + r^3\sum_{n=1}^{M}\varpi_n Q_n)$

of SPRB-TRPCA is ζ'. Suppose that $\kappa_1 = \max(r, c)$, $\vartheta_1 = \min(r, c), \kappa_2 = \max(r, c, b)$, $\vartheta_2 = \min(r, c, b), m = \max(m_v), N_{(1)} = \max(m, \delta), N_{(2)} = \max(n_{(2)}), n_{(1)} = \max(m_v, \delta)$, and $n_{(2)} = \min(m_v, \delta)$ where m_v is the number of spatial pixels within the vth homogenous spatial cluster, and n is set as a factor of the b. For SDe-CNN, K denotes the number of neighboring spectral variables and ζ_i is the iteration time on the ith band. In M-LRTR, p serves as the power index of the Schatten p-norm function. For CRPCA, ζ_1 and ζ_2 are two main parameters of CRPCA with the number of basis vectors k. It should be noted that since some methods do not directly give the computational complexity, the estimation results in Table VI may change as the actual processing is modified and serve as references.

The results in Table VI and Fig. 11 are basically consistent. Specifically, the proposed model operates more efficiently on different HSIs than other global LRTR models but is still slightly inferior to RPCA and SDe-CNN in most cases. Although the corresponding complexity of SS-PLRQR seems to consist of several parts, the calculations are limited to small disjoint local spaces, significantly reducing the computational burden with $m \ll rc, n < b, N_{(1)} < \kappa_1$, and $N_{(2)} < \vartheta_1$. Meanwhile, the actual running time ought to be more significant than the current results of SDe-CNN for relying on specialized software and not accounting for the time needed for network construction. Besides, based on the power

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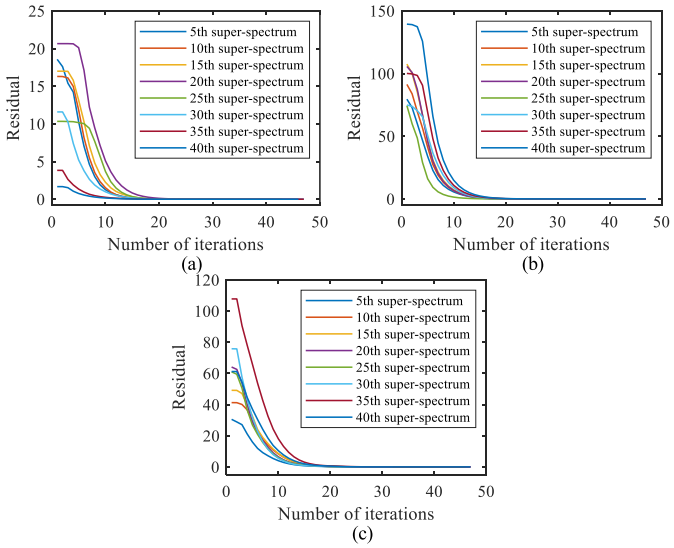


Fig. 12. Convergence curves on different super-spectrums on (a) Indian Pines, (b) Pavia University, and (c) Longkou.

computation in E-LRTR and the multilinear compression of M-LRTR, the corresponding costs are significantly higher than the others. Fortunately, the proposed method's computation time and complexity are acceptable.

G. Convergence Analysis

To verify the convergence of the proposed model, we present the residual curves for the 5th, 10th, 15th, 20th, 25th, 30th, 35th, and 40th super-spectrum obtained through the SS-PLRQR method on the Indian Pines, Pavia University, and Longkou datasets, as shown in Fig. 12. As illustrated in Fig. 12, the super-spectra meet the convergence criteria within 50 iterations across different datasets. Regardless of their initial size, the residual values decline rapidly before the tenth iteration and stabilize close to zero before reaching the maximum number of iterations. This swift convergence is primarily attributable to the parallel operations performed on the super-spectra. Based on these findings, we have set the maximum number of iterations for each subcube at 100, ensuring that the algorithm can achieve convergence across various hyperspectral datasets.

V. Conclusion

This article discusses SS-PLRQR, which combines novel HSSGS with PLRQR. Extensive experiments demonstrate that SS-PLRQR performs excellently in quantitative and qualitative measures, providing valuable insights and useful references for further practical research. SS-PLRQR is a feature extraction method and a variant of traditional low-rank recovery models. It effectively retains key features while minimizing the impact of various noises in classification tasks although it cannot ultimately preserve the original spectral meanings of the input HSIs.

In the future, the authors plan to optimize the associated algorithms to enhance the model's versatility across different scenarios. Meanwhile, this model may function as an adequate data preprocessing technology to assist in distinct detection analyses of HSIs, promoting efficient performance with significant spatial and spectral homogeneity. It is worth noting that setting super homogeneous pixel patches at a scale of 2 in acquiring super-spectrums is anticipated to address the limitations of sample category labels of various available data. It is supposed to be widely applicable in multidimensional remote sensing datasets that lack category labels, particularly in specific regions that are challenging for humans and tools to access due to geographical and technological constraints.

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